

Technology Stack

Date	27 June 2026
Team ID	
Project Name	Rising Waters-ML Flood Prediction System
Maximum Marks	2 Marks

Technology Stack

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript	Provides a simple and interactive web interface for users to enter weather parameters and view flood prediction results.
2	Backend / Server-Side	Python, Flask Framework	Handles user requests, validates inputs, applies feature scaling, loads the trained XGBoost model, and returns flood prediction results through the Flask web application.
3	Database / Data Storage	Dataset File (Excel/CSV), Joblib Model Storage (floods.save and transform.save)	Stores the historical flood dataset used for model training and saves the trained XGBoost model and StandardScaler for future predictions. No database is used in the current implementation.
4	Cloud / Hosting / Deployment	Render Cloud Platform	Used to deploy the Flask web application publicly and provide online access through a hosted URL.
5	Version Control & CI/CD	Git and GitHub	Used for source code management, version tracking, and collaboration during development.
6	Third-Party APIs / Other Tools	Jupyter Notebook, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, XGBoost, Joblib, Unicorn	Used for data analysis, visualization, preprocessing, machine learning model training, evaluation, model saving, and production deployment.